

business. Aeronautical engineers will find jobs in all these sectors."

To give a very specific example, China has witnessed tremendous economic growth and the aviation industry is booming there; we can expect a parallel growth in India also. There is now made-in-China Airbus A320 airplane, which is being used by Chinese airlines. Estimates are that China will be buying about 150 aircrafts a year for the next 20 years or so. The global production of aircrafts in this class is 700-800 per year, so this 150 represents a significant global market. We can expect a somewhat similar development in India over the next 10-20 years.

The present offset business (when any Indian airline buys an aircraft from

Amongst Indian Institutes of Technology (IITs), the four at Madras, Mumbai, Kanpur and Kharagpur offer courses in aerospace engineering. The Department of Aerospace Engineering at each of these IITs offers four-year B.Tech, five-year dual degree (B.Tech and M.Tech), two-year M.Tech, MS and PhD degrees.

Besides these four, Punjab Engineering College at Chandigarh and Madras Institute of Technology offer various courses in aerospace engineering. A good number of private engineering colleges too have come up across the country, namely, Amity Institute of Aerospace Engineering Research and Studies (UP), IIAEIT (Pune), Indian Institute of Aeronautical

“Design and analysis jobs involve abstract modelling and complex calculations, so one needs to be very comfortable with those—Dr P. Sriram, professor and head, department of aerospace engineering, IIT Madras



Boeing, Airbus etc, the aircraft vendor is required to outsource some fraction of the purchase value from India) is expected to grow and smoothly merge into an Airbus China type activity. With more planes in the sky, more operations and maintenance will be required. So these areas will also grow and become major employers.

However, most of the aerospace-related companies and research organisations are still in the public sector.

How to get started?

For a career in aerospace engineering/aeronautics, one requires four to seven years of college study following high school. To enter this field, the minimum requirement is a four-year bachelor's degree.

"BE/B.Tech, ME/M.Tech, MS and PhD programmes are available at about 75 institutions across the country—almost all in the private sector, with only about half a dozen in the government sector," says Prof. Sriram.

Engineering (IIAE) at Dehradun, SRM University (Chennai), VSM Institute of Aerospace Engineering & Technology (Bengaluru), Sardar Vallabhbhai Patel Institute of Technology (Gujarat), Hindustan Institute of Technology & Science (Chennai) and Noorul Islam University (Tamil Nadu).

The Indira Gandhi National Open University (IGNOU) has also started a B.Tech programme in aeronautical engineering in 'on campus' mode. Pune-based Aeronautical Engineering and Research Organisation (AERO) has developed course curriculum to facilitate the programme.

Coming to the fee structure, IITs and other government colleges charge Rs 200,000-300,000 for the four-year course, while private institutions may charge upward of Rs 500,000. At IITs, this includes hostel accommodation and mess charges. Indian Institute of Space Technology, Thiruvananthapuram, does not charge any fees for a course in aerospace engineering; even

messing is free.

"All five-year dual-degree students and two-year M.Tech students get MHRD scholarship of Rs 8000 per month. Some of them get other sponsored scholarships (from Boeing, GE, Honeywell, etc) of higher values. All doctoral students get scholarship to the tune of Rs 14,000 per month for the first two years and Rs 15,000 per month for the next two years (MHRD recently announced an increase in these figures to Rs 18,000 and Rs 20,000, respectively). Most of the undergraduate students also get either state or national or various institute scholarships. There are many endowment and sponsored scholarships (from DRDO, Boeing, etc) for the undergraduates as well. Many of these scholarships are about two-three times the yearly expenses of the students," says Prof. Sinhamahapatra of IIT Kharagpur.

The big picture: know your role

The aerospace industry provides you with a number of career options from engineers, scientists and technicians upstream to other downstream business management roles—whereby you can pick a job role matching your area of interest. Most of the roles, however, can be grouped under some broad categories: managers, analysts, consultants, scientists, field engineers and design engineers.

Those who are interested in R&D can design a satellite launcher, test the latest environment-friendly turbine or create the latest fighter plane as part of the production team. You may also consider joining the human resources team of a large aerospace manufacturer to develop strategies for attracting, recruiting and retaining skilled employees.

During the design process of an aircraft, the aerodynamicist works hard with other design engineers to make sure that the airplane moves easily through the air. The design engineer, on the other hand, decides how long a plane has to be to hold a certain num-